

A Primer on Permutation Tests

bootstrap and shuffling

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Permutation tests: the most general idea

Structure



Random Pattern



In almost all hypothesis testing scenarios, the null hypothesis can be interpreted as follows:

H0: Any pattern that has been seen in the sampled data is simply due to random chance.

<https://byuistats.github.io/Statistics-Notebook/PermutationTests.html>

why we use a permutation test?

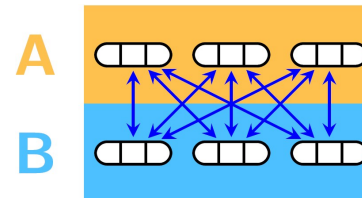
- sometimes precise distributions are not known
- a permutation test makes weaker assumptions about distributions than parametric tests
- permutation tests provide exact inference
- permutation testing applies in the same way to uni- and multivariate analysis

Today:

- No recipes, but the **basic principles** underlying permutation tests
- Some examples

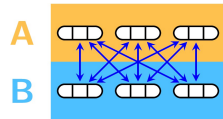
Two main approaches to perform permutation tests

Permutation (shuffling) methods use sampling without replacement to test hypotheses of 'no effect';

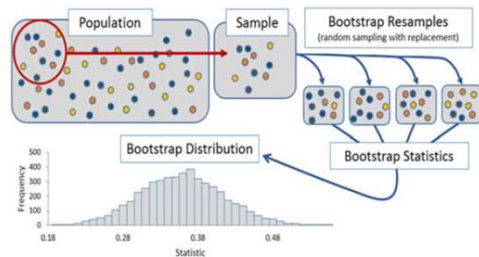


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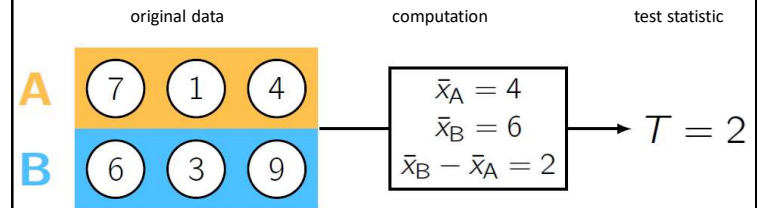


Bootstrap methods use sampling with replacement to establish confidence intervals;



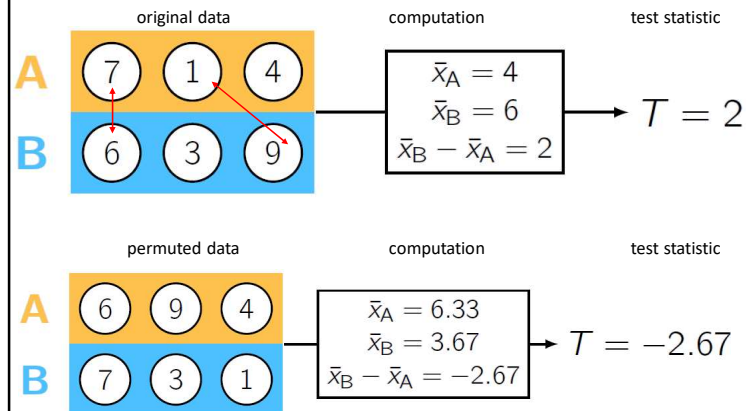
Univariate example: two-sample test

is there a mean difference between groups A and B?



Univariate example: two-sample test

if not (H_0), values can be **exchanged** between conditions

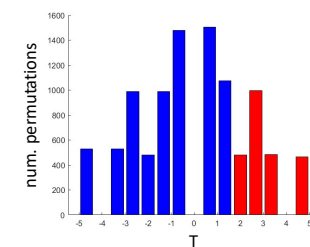


Univariate example: two-sample test

compute the test statistic for many permutations

perm	data		T
1	7 1 4	6 3 9	2.00
2	7 1 6	4 3 9	0.67
3	7 1 3	4 6 9	2.67
4	7 1 9	4 6 3	-1.33
...	1 4 3	7 6 9	4.67
N	4 3 9	7 1 6	-0.67

Draw a histogram



$$p\text{-value} = \frac{N_{red}}{N_{red} + N_{blue}}$$

formal procedure for a permutation test

- 1) compute test statistic T_1 for original data
- 2) compute test statistic T_i for all permutations $i = 2 \dots nP$; $i = 1$ is the 'neutral permutation'
- 3) determine the rank of T_1 : $r = \sum_{i=1}^{nP} [T_i \geq T_1]$ where $[true] = 1$ and $[false] = 0$
- 4) determine the p-value: $p = \frac{r}{nP}$

the test is exact if significance level is a multiple of $1/nP$ and there are no ties, otherwise it is conservative

Example: t-test failure

Gene expression

WT mice	KO mice
1.418786	2.302301
2.312901	2.263253
1.986329	2.474145
0.210846	1.188719
1.646258	2.171384
1.449331	2.284808
1.346526	1.828237
1.323932	1.141976
1.603654	0.864154
2.982957	3.26138

The Four Assumptions Made in a t-Test

Independence: The observations in one sample are independent of the observations in the other sample.

Random Sampling: Both samples were obtained using a random sampling method.

Normality: Both samples are approximately normally distributed.

Homogeneity of Variances: Both samples have approximately the same variance.

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	WT	KO
KS; p-value	0.5528	0.7257
Lilliefors; p-value	0.1138	0.2759

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two-sample F-test p-value: 0.9739

KS: p-value = 0.7625
Lilliefors: p-value = 0.1380.2759

PASSED

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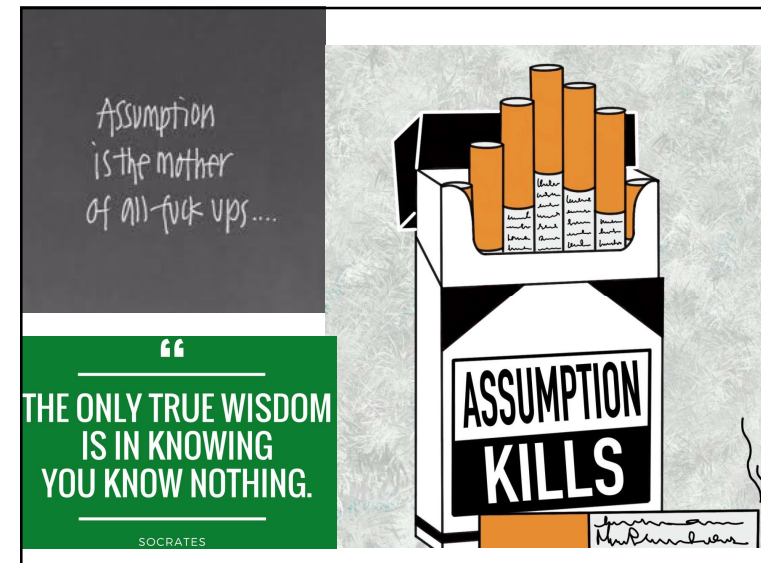
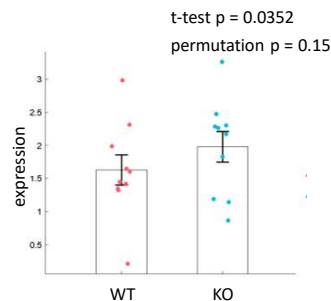
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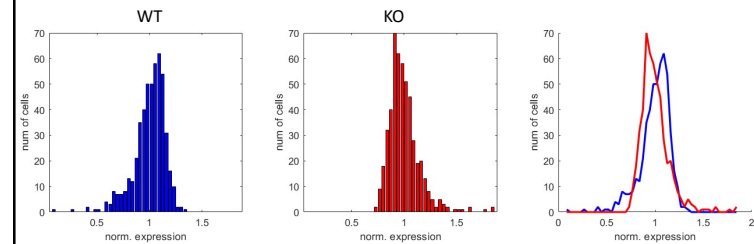


Source of inconsistency between permutation and parametric statistics

Inconsistent results can most often be caused by outliers. Our case.

In general, parametric models give wrong results if the data are not consistent with the model's constraints. For example, the data are generated from a non-normal distribution.

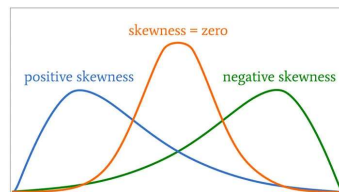
Permutational tests for “self-made” statistics



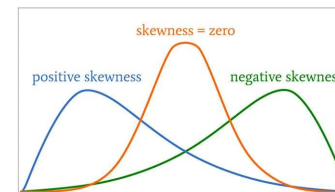
Mean and SD are equal.

What about skewness???

Permutational tests for “self-made” statistics



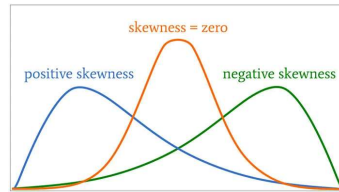
Permutational tests for “self-made” statistics



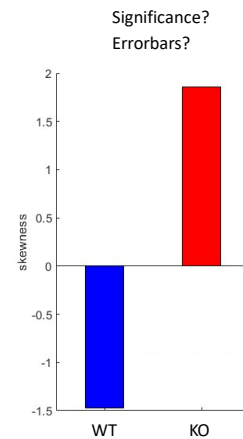
No good “parametric” tests for skewness

May be easily done via shuffling

Permutational tests for “self-made” statistics



No good “parametric” tests for skewness
May be easily done via shuffling



Permutational tests for “self-made” statistics

Original data
“Neutral” permutation

WT	KO
0.5916	0.8326
1.0927	1.0969
...	...
...	...
1.0817	0.8459
0.9374	0.9105

Skew Δ Skew₁

-1.47	1.86
3.33	

Permutation 1

WT	KO
0.8326	0.5916
1.0927	1.0969
...	...
...	...
0.8459	1.0817
0.9374	0.9105

Skew Δ Skew₂

-0.07	0.28
0.35	

Permutation 2

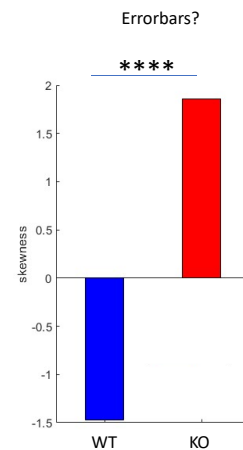
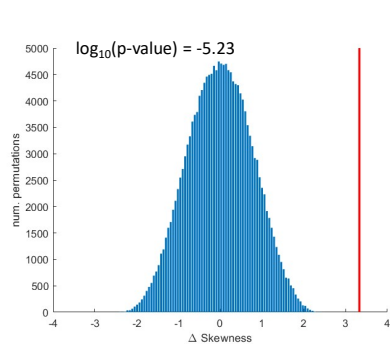
WT	KO
0.5916	0.8326
1.0969	1.0927
...	...
...	...
1.0817	0.8459
0.9105	0.9374

Skew Δ Skew_n

0.46	-0.25
-0.71	

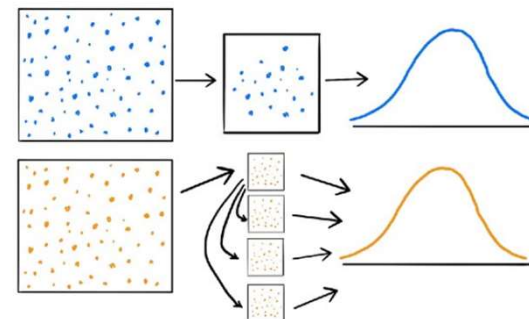
Pool all Δ Skew in one variable – H₀ distribution

Permutational tests for “self-made” statistics

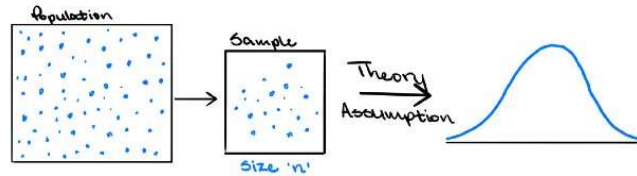


Bootstrapping Statistics. What it is and why it's used.

Bootstrap methods use sampling with replacement to establish confidence intervals



Classic approach



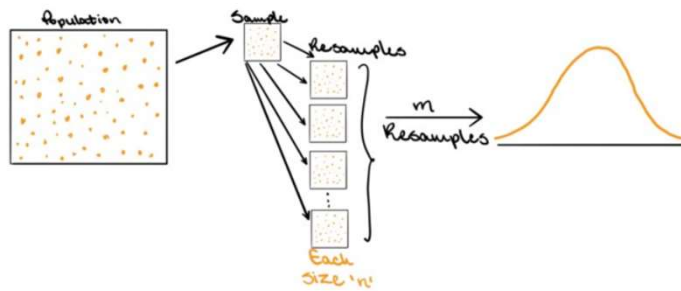
Most times when conducting research, it is impractical or impossible to collect data from the WHOLE population.

Instead, a subset of the population is taken, and insight is gathered from that subset to learn more about the population.

Bootstrap approach

- Bootstrapping is a statistical procedure that resamples a single dataset to create many simulated samples.
- This process allows you to calculate standard errors, construct confidence intervals, and perform hypothesis testing for numerous types of sample statistics.
- Bootstrap methods are alternative approaches to traditional hypothesis testing and are notable for being easier to understand and valid for more conditions.

Bootstrap approach



- As with the traditional approach, a sample of size n is drawn from the population within the bootstrapping approach.
- The sampling distribution is created by resampling observations with replacement
- The benefit of more resamples, then, is to derive a better estimate of the sampling distribution.

Bootstrap to calculate CI for skewness example

Original data
"Neutral" resampling

WT	KO
Cell w1	Cell k1
Cell w2	Cell k2
...	...
Cell w3	Cell k3
Cell w4	Cell k4
$Skew_1^{wt}$	$Skew_1^{ko}$

Resampling 1

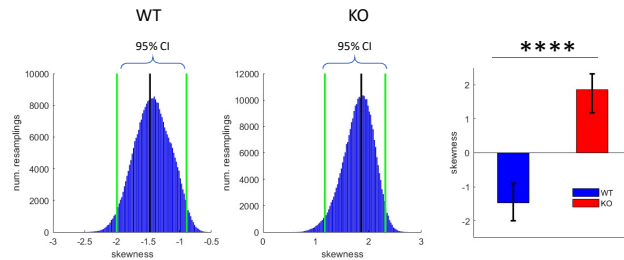
WT	KO
Cell w2	Cell k1
Cell w2	Cell k1
...	...
Cell w3	Cell k9
Cell w4	Cell k9
$Skew_2^{wt}$	$Skew_2^{ko}$

Resampling 2

WT	KO
Cell w1	Cell k3
Cell w2	Cell k4
...	...
Cell w5	Cell k7
Cell w5	Cell k8
$Skew_3^{wt}$	$Skew_3^{ko}$

Pool all $Skew^{wt}$ and $Skew^{ko}$ to get sample distributions

Bootstrap to calculate CI for skewness example



Application of the hierarchical bootstrap to multi-level data in neuroscience



Published in final edited form as:
Neuron Behav Data Anal Theory. 2020 ; 3(5): .

Application of the hierarchical bootstrap to multi-level data in neuroscience

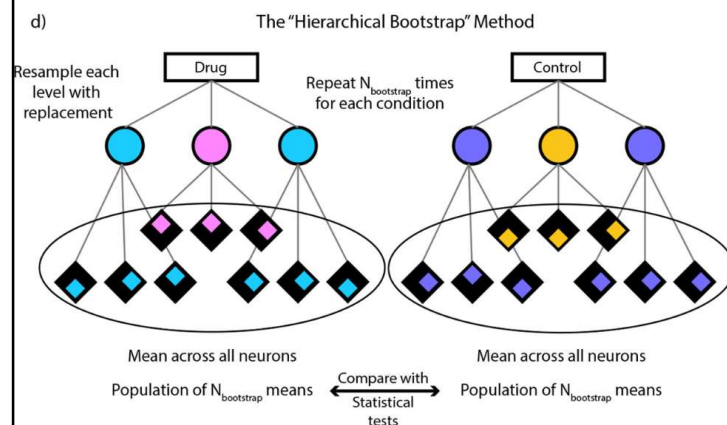
Varun Saravanan¹, Gordon J Berman^{2,3}, Samuel J Sober²

¹Neuroscience Graduate Program, Graduate Division of Biological and Biomedical Sciences, Laney Graduate School, Emory University, 30322

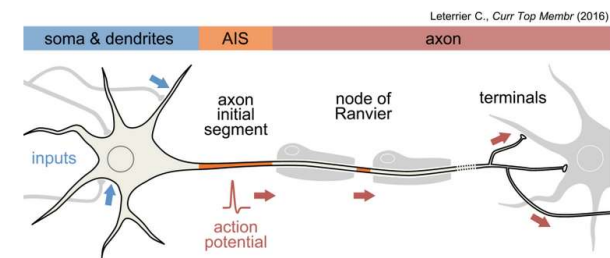
²Department of Biology, Emory University, 30322

³Department of Physics, Emory University, 30322

Application of the hierarchical bootstrap to multi-level data in neuroscience



Permutation tests for the analysis of nonlinear relationships



Matrix Biology 136 (2025) 22–35

Contents lists available at ScienceDirect

Matrix Biology

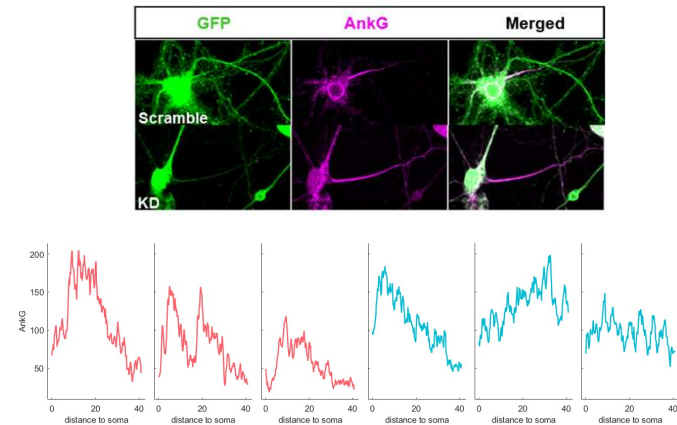
journal homepage: www.elsevier.com/locate/matbio

Neurocan regulates axon initial segment organization and neuronal activity

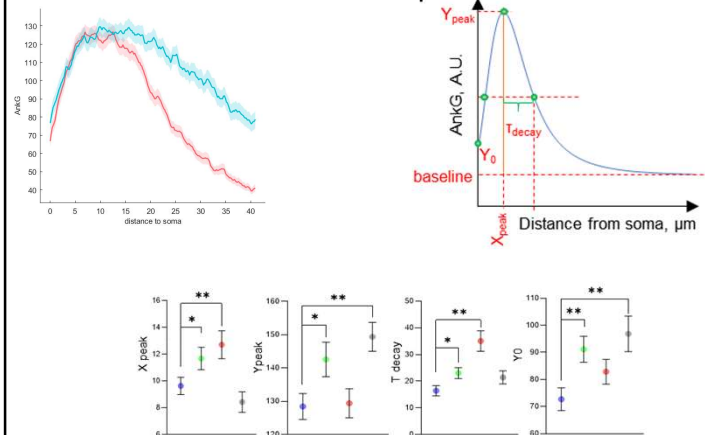
David Baidoe-Ansah^{a,1}, Hadi Mirzapourdelavar^{a,1}, Stepan Aleshin^a,
Björn Hendrik Schott^{b,c,d,e}, Constanze Seidenbecher^{b,c,d,e}, Rahul Kaushik^a,
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^b Leibniz Institute for Neurobiology (LIN), Magdeburg, Germany
^c Center for Behavioral Brain Sciences (CBBS), Magdeburg, Germany
^d Department of Psychiatry and Psychotherapy, University Medical Center, Göttingen, Germany
^e German Center for Neurodegenerative Diseases (DZNE), Göttingen, Germany
^f Center for Intervention and Research on Adaptive and Maladaptive Brain Circuits Underlying Mental Health (C-I-R-C), Jena-Magdeburg-Halle, Germany
^g Medical Faculty, Otto-von-Guericke University, Magdeburg, Germany

Permutation tests for the analysis of nonlinear relationships



Permutation tests for the analysis of nonlinear relationships



Permutation tests for the analysis of cumulative distributions (KS-test)

Cellular and Molecular Life Sciences (2023) 80:82
<https://doi.org/10.1007/s00018-023-04720-z>

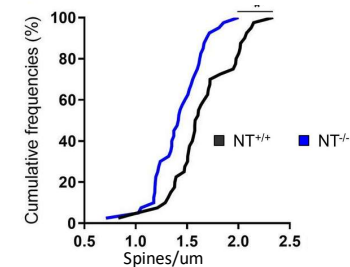
Cellular and Molecular Life Sciences

ORIGINAL ARTICLE

Check for updates

Mice deficient in synaptic protease neurotrypsin show impaired spaced long-term potentiation and blunted learning-induced modulation of dendritic spines

Maura Ferrer-Ferrer¹, Shaobo Jia¹, Rahul Kaushik¹, Jenny Schneeberg¹, Izabela Figiel², Stepan Aleshin¹,
Andrey Mironov¹, Motahareh Safari¹, Renato Frischknecht¹, Jakub Wlodarczyk², Oleg Senkov¹,
Alexander Dityatev^{1,4,5}



Permutation tests for the analysis of cumulative distributions (KS-test)

